Vulnerability-oriented Testing for RESTful APIs

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Vulnerability-oriented Testing for RESTful APIs

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With the increasing popularity of APIs, ensuring their security has become a crucial concern. However, existing security testing methods for RESTful APIs usually lack targeted approaches to identify and detect security vulnerabilities. In this paper, we propose VOAPI2, a vulnerability-oriented API inspection framework designed to directly expose vulnerabilities in RESTful APIs, based on our observation that the type of vulnerability hidden in an API interface is strongly associated with its functionality. By leveraging this insight, we first track commonly used strings as keywords to identify APIs' functionality. Then, we generate a stateful and suitable request sequence to inspect the candidate API function within a targeted payload. Finally, we verify whether vulnerabilities exist or not through feedback-based testing. Our experiments on real-world APIs demonstrate the effectiveness of our approach, with significant improvements in vulnerability detection compared to state-of-the-art methods. VOAPI2 discovered 7 zero-day and 19 disclosed bugs on seven real-world RESTful APIs, and 23 of them have been assigned CVE IDs. Our findings highlight the importance of considering APIs' functionality when discovering their bugs, and our method provides a practical and efficient solution for securing RESTful APIs.

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